

Physics-Coupled Frequency Dynamic Adaptation Network for Domain Generalized Underwater Object Detection

Linxuan Luo
College of Computer Science and
Technology, Zhejiang University of
Technology
Hangzhou, China
211124120023@zjut.edu.cn

Pan Mu*

College of Computer Science and
Technology, Zhejiang University of
Technology
Hangzhou, China
panmu@zjut.edu.cn

Cong Bai
College of Computer Science and
Technology, Zhejiang University of
Technology
Hangzhou, China
congbai@zjut.edu.cn

Abstract

Despite technological progress in underwater object detection, there are still problems such as the domain shift stemming from diverse environmental conditions and severe image degradation in underwater environments. To address these challenges, we propose a hypernetwork-powered domain generalization framework that synergizes physical priors with multi-frequency feature learning, called Hy-UOD. To combat domain shift challenges, we devise a meta-learning empowered hypernetwork architecture that synthesizes domain-generalization parameters through environment-specific physical descriptor encoding for cross-domains. To further mitigate the impact of complex degradation on object detection performance, we designed a Multi-frequency Feature Dynamic Adaptation (i.e., MFDA) module based on hypernetwork features and domain-specific information. This module implements a systematic compensation for degraded features through a multi-level dynamic adaptation mechanism: "low-frequency correction, high-frequency refinement, and mid-frequency reconstruction". Experiments on multiple underwater datasets demonstrate the robust detection performance and strong cross-domain generalization capability of our method. The source code will be available at <https://github.com/White-cat-ed/HyUOD>.

CCS Concepts

• Computing methodologies → Object detection.

Keywords

Deep Learning; Underwater Object Detection; Domain Generalized Underwater Object Detection; Hypernetwork

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*Corresponding author.

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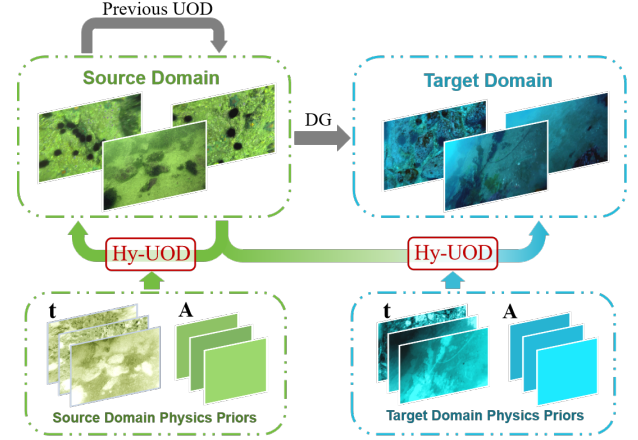


Figure 1: The left and right sides respectively display different domains formed by degradation changes. Previous UOD: Previous underwater object detection methods struggled to generalize to unseen target domains. DG: Conventional domain generalized methods not optimized for underwater environments. Hy-UOD (Ours): By dynamically fusing environment-specific physical descriptor, robust object detection has been achieved in both the source and target domains.

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1 Introduction

In recent years, deep-sea exploration and environmental monitoring have gained increasing attention [18, 23]. Underwater Object Detection (UOD) has emerged as a critical technological component in achieving these objectives. Although terrestrial object detectors [4, 51, 54] have made significant progress by leveraging large-scale datasets such as Microsoft COCO [30] and Open Images [26], their performance significantly declines when applied to underwater environments due to color distortion, noise, and blurring caused by suspended particles [25, 46, 49].

To address the challenges posed by underwater environments, existing UOD methods can be broadly divided into two categories: image enhancement-based methods [7, 10, 13, 19, 45] and feature representation capability-based methods. The former employs image enhancement preprocessing techniques [36, 38, 39, 53] to eliminate interference from underwater image degradation. For example, [10, 35, 45] combine the Retinex theory and dark channel prior to improve detection performance. However, these methods have

limitations: they inevitably lose original image information and misalign with UOD task requirements [18, 33, 35, 48]. In contrast, feature representation capability-based methods directly address feature learning by designing feature extraction mechanisms tailored. For instance, solutions have been proposed based on image blur characteristics [11, 43] and underwater noise issues [55], respectively. However, the limitation of these methods lies in their insufficient consideration of domain shifts, resulting in a sharp decline in the model's generalization in unknown scenarios [6, 9, 32, 37]. To address underwater domain shifts, several domain generalization UOD methods have been proposed [6, 9, 32]. These methods focus on learning transferable feature representations from the source domain, ensuring robust detection performance in unknown underwater environments without requiring additional fine-tuning. Nevertheless, the static architectures of current approaches constrain their ability to detect unknown environments, and their failure to account for intricate underwater degradation processes further weakens their domain generalization performance.

Overall, despite technological progress in underwater object detection, two critical issues persist: (1) significant detection performance degradation in **turbid underwater environments**. For example, the selective absorption and scattering of light by water and variations in turbidity due to changes in suspended particle concentrations, make it difficult to extract the object features under the conditions of reduced contrast and blurred details; (2) **domain adaptation challenges** caused by varying water quality conditions. For example, salinity and temperature variations affect water's refractive index and dissolved organic matter also alters water's optical properties, complicating the adaptation of detection models across different underwater areas.

Contribution: To overcome the limitations aforementioned, this work proposes a hypernetwork-powered domain generalization UOD framework (i.e., Hy-UOD) that deeply integrates environment-specific physical descriptors with a multi-frequency dynamic adaptation structure. As shown in Figure 1, our method is the first to simultaneously achieve robust object detection performance and strong cross-domain generalization capability in severely degraded underwater environments. **To deal with domain generalization challenge**, we extract two key environment-specific physical descriptors from degraded images: the global background light A (representing color cast) and the transmission map t (representing turbidity). To construct feature representations with strong generalization capabilities, we innovatively design the Atmospheric-light Hypernetwork Branch (i.e., AHB), and Transmission-map Hypernetwork Branch (i.e., THB). These meta-learning empowered hypernetworks dynamically generate adaptive parameters based on t and A , thereby guiding the object detection network to perform adaptive feature extraction for different underwater scenarios, ultimately forming a feature representation space with dynamic physical constraints. **To further deal with turbid underwater environments**, we first design a Multi-frequency Feature Dynamic Adaptation module (i.e., MFDA) based on hypernetwork features. This module achieves systematic compensation for degraded features through a multi-level dynamic mechanism: "low-frequency correction, high-frequency refinement, and mid-frequency reconstruction." Specifically, in the low-frequency correction stage, we use AHB driven global dynamic convolution kernels to guide the

model for low-frequency feature correction, effectively eliminating the distortion of low-frequency features caused by hue shifts. In the high-frequency refinement process, by organically combining wavelet decomposition with depthwise convolution, the module effectively suppresses high-frequency noise in images and further extracting multi-scale features of objects. In the mid-frequency reconstruction stage, we innovatively propose a DCP-like feature reconstruction process. This process establishes a latent mapping relationship between the feature space and turbidity representation (which incorporates scene depth structure information) through THB-guided local dynamic convolution.

In summary, our contributions are listed as follows:

- We propose a novel hypernetwork-powered domain generalization underwater object detection framework, Hy-UOD, which deeply integrates environment-specific physical descriptors with multi-frequency dynamic adaptation. the framework achieves robust detection performance in severely degraded environments and demonstrates exceptional cross-domain generalization capabilities.
- We designed the Atmospheric-light Hypernetwork Branch (AHB) and the Transmission-map Hypernetwork Branch (THB). These components guide adaptive feature extraction for diverse underwater scenarios, addressing the domain shift challenge.
- We designed a Multi-frequency Feature Dynamic Adaptation (MFDA) module based on hypernetwork features. Through a three-stage dynamic purification mechanism, it achieves systematic compensation for degraded features.
- We conducted extensive experiments on DUO, RUOD, and URPC2020 for underwater object detection, as well as on S-UODAC2020 for underwater domain generalization object detection. The experimental results demonstrate that our method significantly outperforms state-of-the-art approaches in multiple settings.

2 Related Works

2.1 Underwater Object Detection

UOD method based on image enhancement. The UOD method based on image enhancement mitigates the impact of image degradation on detectors by employing underwater image enhancement preprocessing. For example, [7] proposes a perceptual enhancement model for UOD tasks, incorporating modules that generate visually friendly and detection-friendly images to improve the model's generalization capability in underwater scenarios. [45] presents a reinforcement learning approach that optimizes underwater images specifically for UOD performance rather than visual appeal. GCC-Net [10] enhances UOD performance by fusing raw and enhanced underwater images. However, such methods still have inherent limitations: the enhancement process inevitably causes information loss, and the optimization objectives of enhancement methods are inherently misaligned with those of UOD.

UOD method based on feature representation capability. This type of method aims to optimize feature extraction for underwater degraded images. For example, UDMDet [50] employs a distraction-sensitive FPN and a task-aligned head to enhance feature recognition in underwater images. ERL-Net [11] introduces an

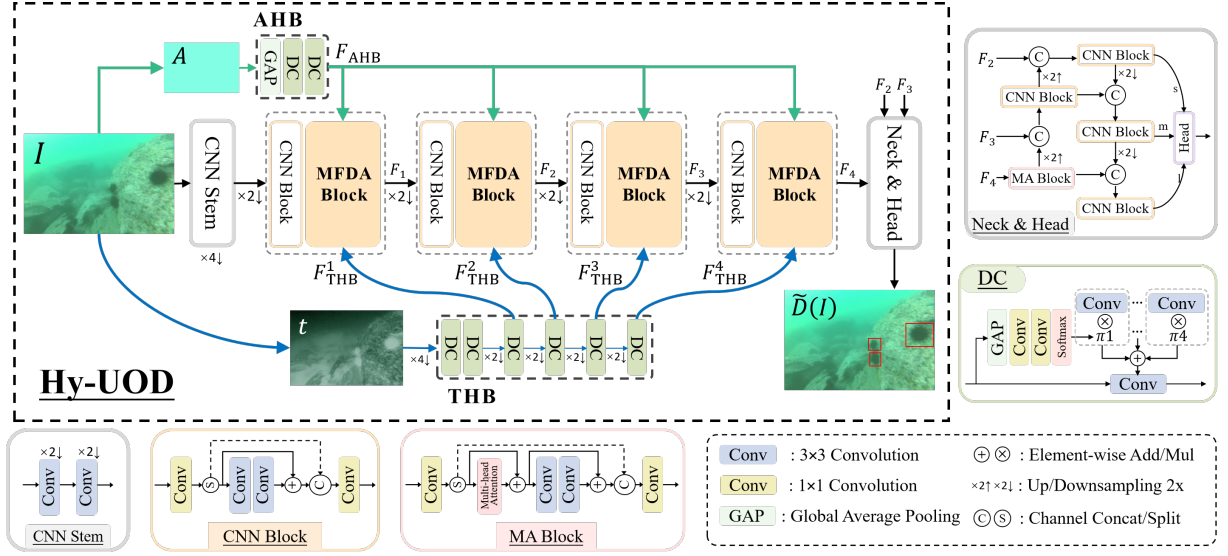


Figure 2: Overall illustration of the proposed Hy-UOD framework. Hy-UOD is composed of three components: Atmospheric-light Hypernetwork Branch (AHB), Transmission-map Hypernetwork Branch (THB) and main object detection network.

edge-guided attention module to model boundary information and a hierarchical feature aggregation module to integrate multi-scale features. AMSP-UOD [55] utilizes amplitude-modulated stochastic perturbations to disrupt noise patterns and vortex convolutions to improve feature extraction.

However, these methods overlook important variations between different water environments. In addition, they neglect underwater physical priors embedded with domain knowledge, which further restricts their performance in handling complex underwater degradation.

2.2 Domain Generalization Underwater Object Detection

The goal of such methods is to learn generalized feature representations from the source domain, enabling stable object detection in unknown domains (new underwater environments) without requiring additional adjustments. For example, DG-YOLO [32] introduces the WQT data augmentation method to diversify underwater domain representations, along with a domain discriminator for cross-domain feature alignment. DMCL [9] employs a CBST underwater network to generate supplementary source domains and performs feature-level mixing to enhance domain diversity. SPIR-DFM [6] uses a domain information reconstruction decoder to reconstruct input images, thereby promoting domain-invariant feature learning. However, existing methods have two key limitations: (1) their static architectures restrict cross-domain generalization ability, and (2) their neglect of complex underwater degradation reduces the effectiveness of domain generalization strategies.

3 Proposed method

3.1 Motivation and Problem Formulation

For the underwater object detection task, this paper aims to construct a domain-generalized detection framework that maintains

robust detection performance in severely degraded environments while demonstrating exceptional cross-domain generalization capabilities. The primary factors causing severe degradation and inter-domain differences are color cast and turbidity changes resulting from underwater optical properties. To solve this problem, we concentrate on enabling the model to learn how to dynamically adjust the network by leveraging environment-specific physical descriptor for domain generalized underwater object detection task. Inspired by the atmospheric scattering model [24], the degraded underwater image can be expressed as

$$I(x) = J(x)t(x) + (1 - t(x))A \quad (1)$$

where x represents the pixel position in the image. $I(x)$ denotes the observed degraded image, while $J(x)$ is the ideal non-degraded image. A is the global background light, $t(x)$ is the transmission map, which quantifies the proportion of light that ultimately reaches the imaging sensor after being scattered and attenuated by water particles. Together, A and $t(x)$ form the environment-specific physical descriptors we require.

With the above descriptors, we designed a meta-learning empowered hypernetwork underwater object detection framework $\mathcal{D}$ (i.e., Hy-UOD), the overall architecture of which is illustrated in Figure 2. Specifically, we developed a Atmospheric-light Hypernetwork Branch $\mathcal{H}_A$ (i.e., AHB), and a Transmission-map Hypernetwork Branch $\mathcal{H}_t$ (i.e., THB). Together, they guide the main object detection network D (i.e., MFDA-Net). Thus, we formulate this problem as follows:

$$\begin{aligned} &\arg \min_{\theta^A, \theta^t} L(\mathcal{D}(I); \hat{y}(I)) \\ &\text{s.t. } \omega \in \arg \min_{\omega} L(\mathcal{D}(I); \hat{y}(I)) \end{aligned} \quad (2)$$

Here, I denotes the image from the training dataset or the source domain dataset, $\hat{y}(I)$ represents its corresponding ground truth, while θ^A , θ^t and ω are the parameters of AHB, THB and MFDA-Net respectively. L indicates the loss function for the object detection

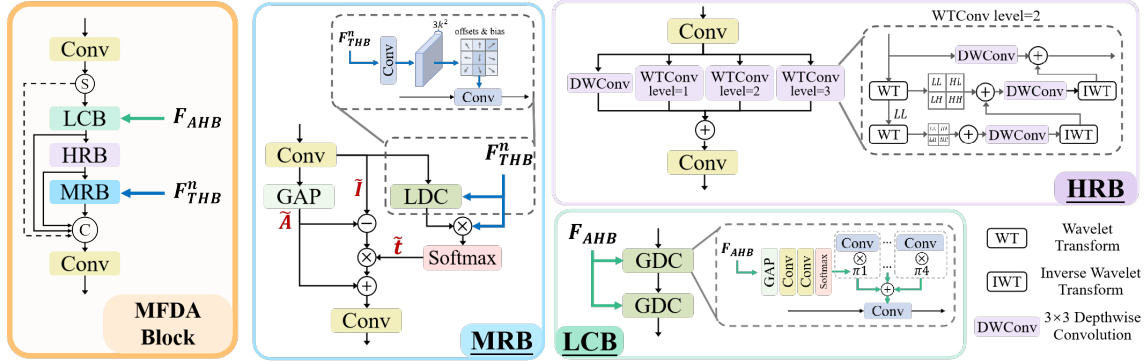


Figure 3: Detailed structure of Multi-frequency Feature Dynamic Adaptation (MFDA) block, including its Low-frequency Correction Block (LCB), High-frequency Refinement Block (HRB), and Mid-frequency Reconstruction Block (MRB) components

task. $\mathcal{D}(\mathbf{I})$ is estimated by the following formula:

$$\mathcal{D}(\mathbf{I}) \leftarrow D(\omega; \mathcal{H}_A(\theta^A, \mathbf{A}(\mathbf{I})), \mathcal{H}_t(\theta^t, \mathbf{t}(\mathbf{I})); \mathbf{I}) \quad (3)$$

3.2 Physics-based Hypernetwork Structure

In this section, we provide a detailed introduction to the proposed Hy-UOD framework. The framework consists of two core components: the meta-learning empowered hypernetwork and the main object detection network.

Meta-learning Empowered Hypernetwork. We first estimate the transmission map $\mathbf{t}(\mathbf{I})$ and the global atmospheric light $\mathbf{A}(\mathbf{I})$ of the input image $\mathbf{I}$ based on the dark channel prior method [15]. Given the heterogeneity in the information representation of $\mathbf{t}(\mathbf{I})$ and $\mathbf{A}(\mathbf{I})$, we design a dual-branch hypernetwork architecture: AHB and THB. Considering that $\mathbf{A}(\mathbf{I})$ primarily carries the hue features (representing global information) of $\mathbf{I}$, The AHB adopts an architecture consisting of a global pooling layer followed by two Dynamic Convolutions (i.e., DC) [8]. This design effectively encodes $\mathbf{A}(\mathbf{I})$ into the hypernetwork features $F_{AHB}(\mathbf{I})$. This process can be formally expressed as:

$$F_{AHB}(\mathbf{I}) = \mathcal{H}_A(\theta^A, \mathbf{A}(\mathbf{I})) \quad (4)$$

Recognizing that the transmission map $\mathbf{t}(\mathbf{I})$ encodes local turbidity characteristics with spatial dependencies, The THB implements a hierarchical architecture. This design sequentially alternates between DC layers and downsampling operations, generating stage-aligned hypernetwork features that provide multi-scale guidance to the main network MFDA-Net. The complete process can be formally represented as:

$$F_{THB}^n(\mathbf{I}) = \mathcal{H}_t(\theta^t, \mathbf{t}(\mathbf{I})), \quad n = 1, 2, 3, 4 \quad (5)$$

Main Object Detection Network. This network is the basic part of object detection. The MFDA-Net employs a four-stage backbone architecture. Initially, a CNN Stem performs basic feature extraction, followed by a hierarchical network comprising alternating CNN Blocks and Multi-frequency Feature Dynamic Adaptation (i.e., MFDA) blocks across four stages. During feature extraction, the MFDA Block progressively integrates hypernetwork features F_{AHB} and F_{THB}^n to reduce the impact of underwater image degradation. In terms of neck design, the network takes the feature maps F_2 , F_3 and F_4 , output from the last three stages of the backbone network, as inputs. First, the MA Block, based on the multi-head

attention [44], is used to model the global context of the high-level features F_4 , enhancing the expressive power of high-level features. Then, inspired by PANet [34], a bidirectional feature pyramid is constructed to achieve effective fusion and interaction of F_2 , F_3 and F_4 . The bottleneck ultimately generates discriminative multi-scale representations for object detection, which the detection head leverages for precise bounding box regression and category classification. As the core innovation, the MFDA module's architecture and mechanism are detailed in Section 3.3.

3.3 Multi-frequency Feature Dynamic Adaptation

In this section, we introduce on the architecture of the MFDA block, as illustrated in Figure 3. Based on hypernetwork features F_{AHB} and F_{THB}^n , this module establishes a multi-level frequency domain dynamic adaptation mechanism of "low-frequency correction, high-frequency refinement, and mid-frequency reconstruction," achieving systematic compensation for degraded features. Consequently, Hy-UOD not only possesses strong cross-domain generalization capabilities but also exhibits robust detection performance against severe underwater degradation.

Low-frequency correction. Since the global atmospheric light $\mathbf{A}(\mathbf{I})$ contains the color-cast information of the degraded image (representing low-frequency degradation), we designed the Low-frequency Correction Block (i.e., LCB), which utilizes the hypernetwork features F_{AHB} to guide the network in correcting low-frequency degradation. Specifically, we employed two global dynamic convolutions. Unlike DC [8], we use F_{AHB} to generate the convolution kernels.

High-frequency refinement. For underwater degraded image, the network architecture's ability to extract high-frequency edge features and fuse multi-scale features is indispensable [11]. Therefore, we propose a multi-scale High-frequency Refinement Block (i.e., HRB) based on WTConv [17], where WTConv achieves efficient multi-scale high-frequency feature processing by organically combining wavelet transforms [12] and 3×3 depthwise convolutions. Specifically, The HRB employs four parallel branches, consisting of one DWConv and three WTConv at different levels, each constructing equivalent receptive fields of $[3 \times 3, 6 \times 6, 12 \times 12, 24 \times 24]$. Finally, the HRB's output is generated by element-wise addition of

the four feature maps followed by a 1×1 convolution. This design not only enhances the model's ability to capture high-frequency details but also maintains high computational efficiency.

Mid-frequency reconstruction. Based on Akkaynak's light scattering model [1], $t(x)$ can be expressed as $t(x) := e^{-\beta d(x)}$, where β is the water attenuation coefficient, and $d(x)$ represents the scene depth at pixel x . This model indicates that there is an exponential decay relationship between the transmission map $t(x)$ and the scene depth $d(x)$, with $d(x)$ inherently characterizing the three-dimensional structure of the underwater scene (i.e., the mid-frequency structural features). Inspired by the Dark Channel Prior (i.e., DCP) [22] for the inverse restoration process of degraded images, we further designed the Mid-frequency Reconstruction Block (i.e., MRB). The MRB establishes a latent mapping relationship between network features and turbidity representation (which contains mid-frequency structural information) through spatially dynamic convolution guided by F_{THB}^n . Furthermore, the MRB achieves a feature reconstruction process similar to DCP at the feature level based on this mapping relationship. Specifically, the image restoration process of DCP can be expressed as:

$$J(x) = t(x)^{-1}(I(x) - A) + A \quad (6)$$

The DCP-like feature reconstruction process after its transformation can be described by the following mathematical expressions:

$$\tilde{J} = \tilde{t}(\tilde{I} - \tilde{A}) + \tilde{A} \quad (7)$$

Among them, $\tilde{I}$ is obtained through a 1×1 convolution, while $\tilde{A}$ is derived by applying global average pooling to $\tilde{I}$. Regarding the generation process of $\tilde{t}$: first, we design a Local Dynamic Convolution (i.e., LDC) based on DCNv3 [47] to constrain and align the feature space of $\tilde{I}$ with F_{THB}^n ; then, the aligned features are element-wise multiplied with F_{THB}^n and normalized via softmax, ultimately producing the attention map $\tilde{t}$ containing transmission map information and mid-frequency structural information.

3.4 Loss Functions and Optimization Procedure

This study employs a combination of localization loss and classification loss to supervise the network training process. Specifically, the localization loss combines Complete Intersection over Union Loss L_{CIoU} with Distribution Focal Loss [28] L_{DFL} , while the classification task is optimized using Binary Cross-Entropy Loss L_{BCE} . The total loss L_{total} of Hy-UOD training can be expressed by the formula:

$$L_{total} = \alpha L_{CIoU} + \beta L_{DFL} + \gamma L_{BCE} \quad (8)$$

where α , β and γ represent hyperparameters that control the weighting of each loss component. In our experimental setup, these hyperparameters were configured as follows: 7.5, 0.5 and 1.5.

4 Experimental Results

This chapter first presents the experimental setup in Section 4.1. Subsequently, Section 4.2 verifies the performance of the proposed detection method in complex, degraded environments, while Section 4.3 demonstrates its cross-domain generalization capability. Additionally, we conduct an ablation analysis on the proposed components and present the corresponding analyses in Section 4.4.

4.1 Experimental Setup

Datasets. This study employed four representative underwater object detection datasets for experimental validation, including three commonly used underwater object detection datasets (DUO [31], RUOD [18], URPC2020) and one underwater domain generalization object detection dataset (S-UODAC2020 [9]). The DUO dataset contains 4 object categories, totaling 7,782 images, with 6,671 used for training and 1,111 for testing. The RUOD dataset covers 10 object categories, comprising 9,800 training images and 4,200 test images. The URPC2020 dataset includes 4 categories, with 5,543 images in the training set and 1,200 images from validation set used for testing. The S-UODAC2020 dataset, serving as a domain generalization dataset, contains 5,542 images across 7 domains. In the experimental setup, we used 4,745 images from type-1 to type-6 as source domain data and 797 images from type-7 as target domain data.

Evaluation Metrics. This study employs the widely recognized COCO [30] metrics to assess model performance, specifically including two metrics: standard average precision (AP) and average precision at an IoU threshold of 0.5 (AP50). Additionally, to evaluate the computational efficiency of the model, we further analyze the number of parameters (Params) and computational complexity (FLOPs).

Implementation Details. Our experiments were conducted on an Intel(R) Xeon(R) Platinum 8358 CPU @ 2.60GHz and an NVIDIA RTX A6000 GPU, running on the Ubuntu 20.04 LTS operating system. The environment was built with Python 3.8.19 using Anaconda, and the network architecture was implemented with PyTorch 2.1.2. The network was trained using the SGD optimizer with random initialization and a learning rate of 0.01. The training was performed over 400 epochs with a batch size of 16, using a random seed of 0. No pre-trained model was utilized.

4.2 Qualitative and Quantitative Comparison

To verify the detection performance of the proposed method in complex, degraded underwater environments, we present the evaluation results under different degrees of degradation datasets, i.e., DUO [31], RUOD [18] and URPC2020, as shown in Table 1. We selected three major categories of methods for comparative experiments: object detection (OD) methods, the YOLO series, and underwater object detection (UOD) methods. Among them, the object detection methods include Faster R-CNN [40], Cascade-rcnn [3], RetinaNet [29], Atss [52], ToD[16], Deformable-detr [56] and Dino [51]; the YOLO series includes YOLOv5, YOLOv6 [27], YOLOx [21], YOLOv8 and YOLO11; and the underwater object detection methods include Boosting R-CNN [43], GCC-Net [10], AMSP-UOD [55], UnitModule [35] and UDMDet [50]. The above comparison methods do not use pre-trained models in the experiments.

Performance on severely degraded environment. As shown in Table 1, the proposed method demonstrates significant advantages in a severely degraded environment, i.e., on the DUO and URPC2020 datasets, with both AP and AP50 metrics substantially surpassing all comparative methods. Particularly noteworthy is that compared to the recently released YOLO11m model (with 20.0M parameters and 68.2GFLOPs computational cost), our method requires only 12.5M parameters (a 37.5% reduction) and 28.1GFLOPs

Table 1: Quantitative comparison on underwater object detection datasets. Our experiments on severely degraded environments (i.e., DUO, URPC2020) and complex category scenarios (i.e., RUOD) achieve the best performance under most settings. AP: AP@[0.5:0.05:0.95], AP50: AP@0.5; Ho/Ec/St/Sc denote AP50 for holothurian, echinus, starfish, and scallop respectively. Bold indicates the highest.

Method	DUO		DUO Categories AP50				RUOD		URPC2020		Params(M)↓	FLOPs(G)↓
	AP↑	AP50↑	Ho↑	Ec↑	Sc↑	St↑	AP↑	AP50↑	AP↑	AP50↑		
OD Methods:												
Faster R-CNN [40] <i>NeurIPS'15</i>	37.2	61.9	56.7	81.2	30.2	81.0	52.1	81.6	19.5	41.3	41.4	208
Cascade-rcnn [3] <i>CVPR'18</i>	48.0	68.4	61.5	86.5	41.0	84.7	55.7	82.9	20.2	41.8	69.2	236
RetinaNet [29] <i>ICCV'17</i>	34.2	56.2	53.2	77.2	24.2	75.4	51.0	81.6	18.0	38.9	36.4	207
Atss [52] <i>CVPR'20</i>	36.8	58.6	37.6	85.1	31.2	80.5	55.0	83.2	21.9	45.8	32.1	203
Tood [16] <i>ICCV'21</i>	44.4	64.4	49.8	87.4	35.3	85.3	53.7	80.5	26.0	51.4	32.0	199
Deformable-detr [56] <i>ICLR'21</i>	57.5	81.1	84.1	88.1	62.0	90.2	57.9	85.3	33.9	65.0	40.1	193
Dino [51] <i>ICLR'23</i>	61.9	81.8	81.7	92.0	61.5	92.0	59.1	83.9	25.6	50.4	47.5	274
Yolo Series:												
Yolov5s	61.8	82.5	84.2	90.1	66.8	89.2	59.2	85.2	32.3	62.0	7.0	7.9
Yolov6s [27]	63.9	84.1	86.7	91.0	68.2	91.5	61.5	85.6	31.8	60.8	17.2	21.9
YoloXs [21]	64.3	85.1	86.9	91.5	69.1	90.4	61.5	86.8	40.1	71.7	8.9	13.3
Yolov8s	66.0	85.0	86.9	91.5	69.1	92.8	61.1	84.0	40.6	71.4	11.1	14.3
Yolo11s	68.5	85.6	89.2	92.7	66.4	93.9	64.0	86.2	40.8	71.2	9.4	21.6
Yolo11m	70.4	85.8	89.4	92.8	67.2	93.7	65.4	86.8	41.5	71.8	20.0	68.2
UOD Methods:												
Boosting R-CNN [43] <i>IJON'23</i>	47.1	67.2	37.0	65.4	25.4	60.6	47.6	81.0	27.4	52.3	47.7	176.1
UDMDet [50] <i>TII'24</i>	47.0	67.6	33.1	66.3	27.1	61.6	47.4	81.4	26.5	51.7	70.1	246.2
GCC-Net [10] <i>PR'24</i>	54.2	77.1	48.4	68.4	32.3	67.7	52.4	81.9	30.8	60.3	36.7	300.8
UnitModule [35] <i>PR'24</i>	61.1	81.2	85.7	90.9	56.2	92.1	54.3	82.2	36.8	68.2	9.0	13.3
AMSP-UOD [55] <i>AAAI'24</i>	64.9	84.5	85.4	92.2	67.9	92.3	61.4	85.3	38.5	68.5	10.3	23.8
Hy-UOD (Ours)	70.8	87.4	90.1	92.9	72.0	94.6	65.1	87.2	41.7	72.4	12.5	28.1

(a 58.8% reduction) in computational cost, while achieving comprehensive superiority in detection performance. On the URPC2020 dataset, our method achieved the most outstanding performance, consistent with its results on the DUO dataset.

Performance on complex underwater scenarios. To further demonstrate the detection performance of the model on complex underwater scenarios, we present the results on RUOD, as shown in Table 1. Since the majority of the data in the RUOD dataset was collected from mainstream search engines including Bing, Google, and Baidu, it exhibits the following characteristics compared to the DUO and URPC2020 datasets: more complex categories, nearly twice the data size of DUO, and lower degradation levels. These features provide a favorable training environment for large-parameter object detection methods. Nevertheless, with fewer parameters, Hy-UOD still surpasses YOLO11m on the key AP50 metric, proving the effectiveness and robustness of our approach. The experimental results demonstrate that Hy-UOD outperforms most comparative methods.

Qualitative Examples. Figure 4 shows comparative qualitative detection results of Hy-UOD, Yolo11s, and AMSP-UOD across multiple datasets including DUO, RUOD, URPC2020, and S-UODAC2020. Since DUO and S-UODAC2020 share the same object categories, we directly applied the model trained on the DUO training set

to conduct extended qualitative experiments on the Type-7 of S-UODAC2020 (i.e., DUO→S-UODAC).

4.3 Domain Generalization Validation

To comprehensively evaluate the domain generalization ability of the models, we conducted experiments on the S-UODAC2020 dataset [9], which is specifically designed for underwater domain generalization. We selected the YOLO series [21, 27], UOD methods [10, 35, 43, 50, 55], and domain generalization object detection (DGOD) methods as comparative approaches. Among them, the DGOD methods include DANN [20], CrossGrad [42], JiGEN [5], DG-YOLO [32], DMCL [9], and SPIR-DFM [6].

Domain generalization verification on S-UODAC2020. Using type-1 to type-6 in the S-UODAC2020 dataset as the source domain training data and type-7 as an independent target domain test set, a zero-shot evaluation was conducted. This cross-domain transfer scheme not only allowed us to further examine the model's generalization performance in unseen underwater domains but also enabled us to evaluate its robustness under extreme degradation conditions. As shown in Table 2, our algorithm achieved the best cross-domain recognition performance. This result fully demonstrates the strong adaptability of the proposed method to complex underwater environmental changes and its excellent generalization characteristics of the domain.

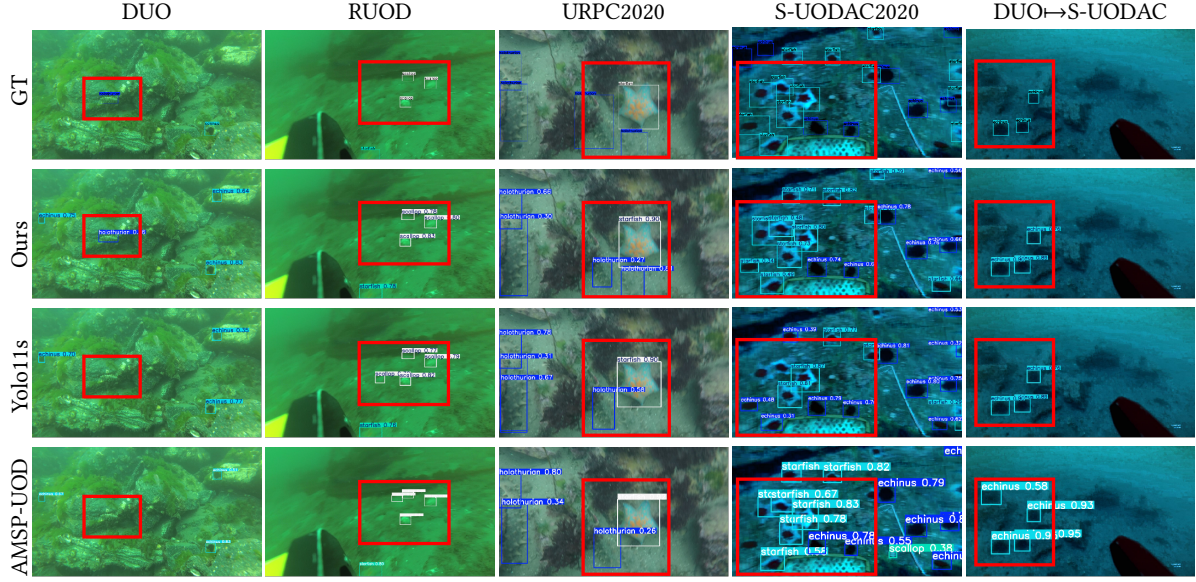


Figure 4: Visualization of qualitative experimental results. Key prediction differences are marked in red.

Table 2: Quantitative comparison on domain generalized underwater object detection dataset S-UODAC2020.

Method	S-UODAC		S-UODAC Categories AP50			
	AP↑	AP50	Ho↑	Ec↑	Sc↑	St↑
Yolo Series:						
Yolov5s	17.8	40.7	29.8	70.7	40.7	21.6
Yolov6s [27]	16.2	38.2	18.6	72.2	35.7	26.3
Yoloxs [21]	23.9	54.7	37.6	71.9	58.5	50.4
Yolov8s	25.5	54.5	36.9	71.5	57.4	53.3
Yolo11s	26.5	54.7	43.6	78.5	55.6	40.2
UOD Methods:						
Boosting R-CNN [43]	13.5	30.6	13.6	57.5	28.7	22.7
UDMDet [50]	13.0	29.2	11.9	57.5	37.1	10.5
GCC-Net [10]	16.5	38.5	21.4	63.3	46.9	22.3
UnitModule [35]	25.5	54.8	35.1	76.9	55.2	52.0
AMSP-UOD [55]	22.8	49.3	31.4	76.9	48.9	40.1
DGOD Methods:						
DANN [20]	-	37.3	25.8	63.7	30.3	24.0
CrossGrad [42]	-	41.8	32.2	71.2	32.5	31.4
JiGEN [5]	-	45.3	34.6	73.2	35.1	38.6
DG-YOLO [32]	-	39.2	32.8	62.7	34.5	26.8
DMCL [9]	-	53.3	43.8	73.3	59.6	36.7
SPIR-DFM [6]	-	54.5	42.1	69.3	54.6	52.3
Hy-UOD (Ours)	28.3	56.7	44.0	80.7	60.9	40.4

4.4 Ablation Study

Comprehensive ablation experiments were conducted on both the DUO and S-UODAC2020 datasets to evaluate the robust detection

performance of our method in complex underwater environments and its cross-domain generalization capability.

Effectiveness of the multi-frequency module. This ablation experiment evaluates the contributions of each component in the MFDA framework in severely degraded environments (i.e., DUO) and domain generalization (i.e., S-UODAC2020). As shown in the lower part of Table 3, the experimental results indicate that the mid-frequency reconstruction component (i.e., $\mathcal{F}_M$) plays a critical role in handling severely degraded underwater environments. its removal leads to significant performance drops on the DUO dataset, with AP decreasing by 1.2% and AP50 by 1.1%. And the low-frequency correction component (i.e., $\mathcal{F}_L$) is essential for domain generalization, as its absence results in noticeable declines on the S-UODAC2020 dataset, reducing AP by 1.1% and AP50 by 1.5%. These results demonstrate that each component in MFDA serves a vital role in underwater object detection.

Effectiveness of the different frequency settings. To show the impact of different component processing sequences in the MFDA module, we evaluated the performance with four settings, as listed in the upper part of Table 3. We tested four processing sequence configurations on the DUO and S-UODAC2020 datasets. The experimental results show that the cascaded sequence (i.e., $\mathcal{F}_L \rightarrow \mathcal{F}_H \rightarrow \mathcal{F}_M$) of low-frequency correction (i.e., $\mathcal{F}_L$), high-frequency refinement (i.e., $\mathcal{F}_H$), and mid-frequency reconstruction (i.e., $\mathcal{F}_M$) maximizes the performance advantages of MFDA, achieving 70.8% AP and 87.4% AP50 on the DUO dataset, and 28.3% AP and 56.5% AP50 on the S-UODAC2020 dataset. The results demonstrate that the sequential "correction-refinement-reconstruction" strategy effectively integrates physical prior knowledge with deep feature learning.

Verification of the effectiveness in severe degraded disturbances. We extract features from the detector neck of the trained model and employ the Grad-CAM [41] method to visualize and analyze these features. As shown in Figure 5, the visualization results on

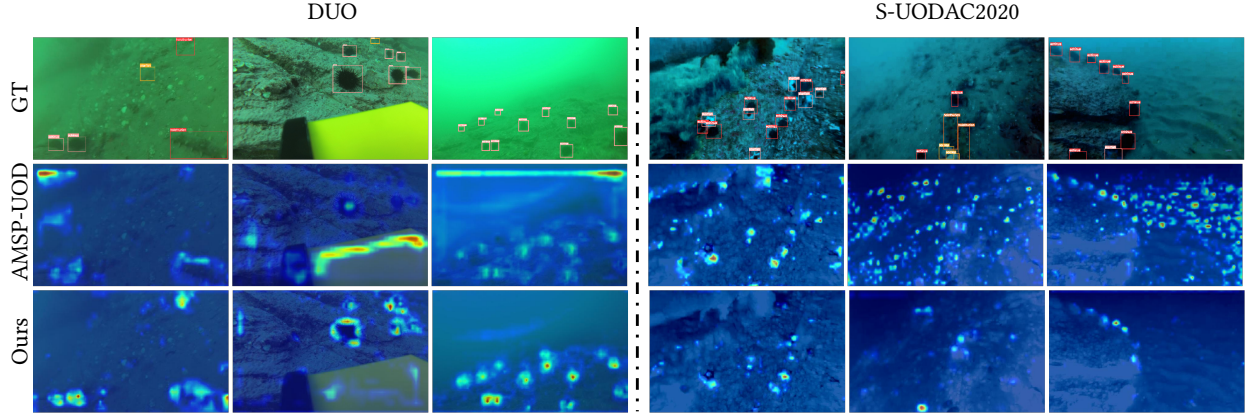


Figure 5: Visualization of attention maps for AMSP-UOD [55] and Hy-UOD (Ours). The attention heatmaps of the detector necks for both models are plotted on underwater images by using the Grad-CAM [41] method.

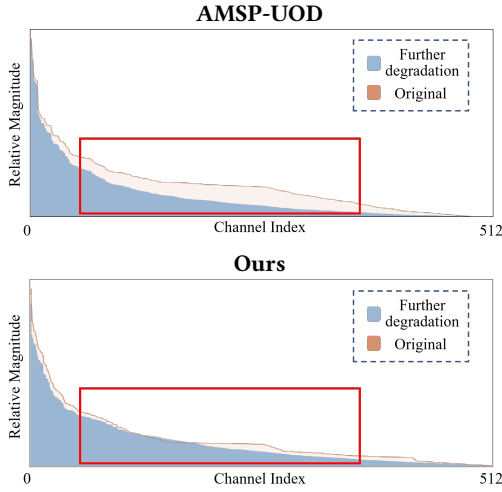


Figure 6: Comparative visualization of feature channel mean magnitudes

the DUO dataset’s test set and the type-7 set of the S-UODAC2020 dataset intuitively demonstrate the performance improvement of our method. It can be observed that AMSP-UOD is prone to interference from regions with varying degrees of degradation in a single image, causing it to focus on meaningless background information in severely degraded areas. In contrast, Hy-UOD can more accurately capture objects in underwater environments, presenting clearer semantic information.

Verification of the robustness. To deeply investigate the impact of degradation on detector performance from the perspective of feature channels, we designed the following experiments. First, based on underwater degraded samples, we synthesized sample pairs with consistent semantic information but different degradation levels using the degradation operations in the UnitModule [35]. Subsequently, on this set of sample pairs, we quantitatively calculated the mean magnitude of each channel [2, 14] in the detector’s neck feature layer. As shown in Figure 6, the results reveal that the feature channel responses of the AMSP-UOD model exhibit

Table 3: Ablation Study of proposed method on DUO and S-UODAC2020.

MFDA				DUO		S-UODAC	
$\mathcal{F}_L$	$\mathcal{F}_M$	$\mathcal{F}_H$	Order	AP $\uparrow$	AP50 $\uparrow$	AP $\uparrow$	AP50 $\uparrow$
✓	✓	✓	$\mathcal{F}_M \rightarrow \mathcal{F}_L \rightarrow \mathcal{F}_H$	70.0	87.3	27.6	55.9
✓	✓	✓	$\mathcal{F}_M \rightarrow \mathcal{F}_H \rightarrow \mathcal{F}_L$	70.2	87.1	27.8	55.7
✓	✓	✓	$\mathcal{F}_L \rightarrow \mathcal{F}_M \rightarrow \mathcal{F}_H$	70.4	86.8	27.9	55.4
✓	✓	✓	$\mathcal{F}_L \rightarrow \mathcal{F}_H \rightarrow \mathcal{F}_M$	70.8	87.4	28.3	56.5
	✓	✓	$\mathcal{F}_H \rightarrow \mathcal{F}_M$	70.1	86.9	27.2	55.0
✓		✓	$\mathcal{F}_L \rightarrow \mathcal{F}_H$	69.6	86.3	27.5	55.2
✓	✓		$\mathcal{F}_L \rightarrow \mathcal{F}_M$	70.4	87.1	27.9	56.1

significant degradation sensitivity. Under further degradation, the responses of medium-intensity channels are suppressed. This phenomenon indicates that AMSP-UOD reduces attention to potential object regions, leading to a decline in recall performance. In contrast, the Hy-UOD framework demonstrate a stable channel response distribution, with the average magnitudes of each channel remaining consistent. This fully validates that the hypernetwork branch can adaptively adjust the feature extraction process based on environment-specific physical descriptor, thereby maintaining the model’s robustness.

5 Conclusion

In this work, we develop a novel hypernetwork-powered domain generalization framework, named Hy-UOD. Our framework constructs the AHB and THB as hypernetworks, leveraging global atmospheric light and transmission maps to dynamically generate parameters for the MFDA module, thus enabling generalization to diverse underwater scenarios. The MFDA module employs a multi-frequency domain adaptation mechanism to decompose and compensate for complex image degradation, thus enhancing detection performance. Experimental results demonstrate that our method achieves superior performance in both quantitative object detection scores and cross-domain generalization capability.

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